

DAVIS PARLOUR

Firmware Engineer | Embedded C | Hardware/Firmware Integration

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PROFESSIONAL SUMMARY

Firmware engineer specialising in embedded software, developing, debugging and maintaining C across four commercial firmware codebases. Take changes from requirements and investigation through design, implementation, source-level debugging, real hardware testing and release. Rapidly pick up unfamiliar systems and domains, taking ownership of difficult engineering problems. Strong in microcontroller firmware, hardware/firmware integration, firmware security, root cause analysis and developer tooling, with a First Class BSc in Computer Science.

KEY SKILLS

Embedded C | Microcontrollers | Firmware design & development | Hardware/firmware integration | Bootloaders | JTAG | Oscilloscope | Source-level debugging | Root cause analysis | Firmware testing & validation | Requirements analysis | SHA-256 | ECDSA | AES-128-CBC | Git | Azure DevOps | CI/CD | Azure Key Vault | VS Code | Ninja | I2C | SPI | UART | AI-assisted engineering

WORK EXPERIENCE

Graduate Firmware Engineer | Palintest (Halma) | Newcastle / Gateshead

September 2025 - Present

- Develop, maintain, test and release embedded firmware in C across four commercial codebases, supporting production products, new features and legacy hardware.
- Own substantial changes end to end: clarify requirements, investigate behaviour, propose the implementation, write and debug firmware, validate on physical instruments and support release.
- Work in a multidisciplinary R&D environment with electronics engineers, using JTAG debugging, a digital oscilloscope and bench instrumentation to inspect signal and interrupt behaviour, reproduce failures, isolate root causes and validate firmware/electrical behaviour.
- Collaborate with chemistry, electronics, sales and marketing teams to translate measurement methods, customer needs and commercial requirements into implementable firmware and product changes.

Selected achievements

- **Build & debugging tooling** - Reduced firmware build time by about 40% by migrating the workflow into VS Code and introducing Ninja; integrated JTAG-based live debugging for breakpoint and source-level investigation without a full rebuild.
- **Compliance & market enablement** - Implemented a standards-driven measurement capability from published technical literature, satisfying compliance and market-access requirements and enabling the product for additional markets.
- **Hardware fault isolation** - Led investigation of abnormal power/interface behaviour on production hardware, reproduced the issue and isolated it to a hardware subsystem; the evidence enabled electronics engineers to identify and fix the component-level root cause.
- **Hardware compatibility** - Implemented firmware compatibility across legacy and updated peripheral variants within a commercial instrument, selecting the correct software path based on the installed hardware.
- **Battery & power integration** - Implemented alkaline/lithium battery support; used a bench power supply and multimeter to simulate conditions, correlate electrical measurements with firmware state and validate behaviour.
- **Secure CI/CD & firmware signing** - Automated firmware signing through Azure DevOps using asymmetric keys protected by Azure Key Vault, keeping private signing material out of source control and normal developer workflows.
- **Firmware security & boot flow** - Worked across bootloader/application flow, implementing application-image signature validation using SHA-256 and ECDSA and protecting sensitive credentials at rest using AES-128-CBC.
- **Product UI & stakeholder communication** - Prototype UI/workflow changes, create raster and vector assets using Affinity Photo and Affinity Designer, and present proposals to stakeholders before implementation.
- **AI tools & enablement** - Selected internally as Palintest's AI Specialist; develop internal AI tools with Azure AI Foundry, including a chatbot that helps senior stakeholders explore and summarise Voice of Customer data.

TECHNICAL SKILLS

Languages: C / Embedded C; C++; Python; Java; Bash; working familiarity with JavaScript, HTML/CSS, PHP and SQL.

Microcontrollers & low-level: Renesas MCUs; bootloader/application flows; GPIO, timers, ADC, PWM, I2C, SPI, UART, watchdogs, sensors and display integration.

Debugging & test: JTAG-based source-level and live debugging, digital oscilloscope signal/interrupt-line analysis, bench power supply, multimeter, structured fault isolation, real hardware testing and validation.

Build & delivery: Git, GitHub, Azure DevOps, CI/CD, Azure Key Vault, VS Code, Ninja, firmware build tooling, software configuration management, firmware signing and release workflows.

EDUCATION

BSc Computer Science - First Class Honours | Lancaster University | 2025

INTERESTS

ITF Taekwon-Do, strength training, running, football, hiking, fishing, tabletop games and coffee.